BIODATA OF PRINCIPAL INVESTIGATOR / KEY PERSONNEL

Standard layout assembled from the master CV; check section order and limits against the call's own template. Data as of 5 November 2026.

A. Personal and professional details

Name	Dr Ramachandran Thiruvengadam
Date of birth	[to be entered by the applicant before submission]
Designation and organisation	Associate Professor, Maternal and Child Health Programme, BRIC-Translational Health Science and Technology Institute (THSTI), an institute under the Biotechnology Research and Innovation Council (BRIC), Department of Biotechnology, Ministry of Science and Technology, Government of India
Address	NCR Biotech Science Cluster, 3rd Milestone, Faridabad-Gurugram Expressway, Faridabad 121001, Haryana, India
E-mail	ramachandran@thsti.res.in
Area of expertise	Maternal and child health; perinatal epidemiology; clinical trials; clinical prediction models, AI-enabled imaging tools and multi-omics for adverse pregnancy outcomes; clinical and translational immunology

B. Qualifications

Degree	Institution	Year
Ph.D. (Life Sciences - Maternal and Child Health)	Translational Health Science and Technology Institute, Faridabad (affiliated to Jawaharlal Nehru University, New Delhi)	2023
M.D. (Biochemistry)	All India Institute of Medical Sciences (AIIMS), New Delhi	2016
Diploma in Child Health (DCH)	Institute of Child Health and Hospital for Children, Madras Medical College, The Tamil Nadu Dr MGR Medical University	2009
M.B.,B.S.	Madurai Medical College, Madurai, The Tamil Nadu Dr MGR Medical University	2005

C. Experience

Position	Organisation	From	To
Associate Professor, Maternal and Child Health Programme	BRIC-Translational Health Science and Technology Institute (THSTI), Faridabad	30 June 2025	Till date
Assistant Professor, Department of Biochemistry	Pondicherry Institute of Medical Sciences, Puducherry	May 2022	June 2025
Principal Scientist - I	THSTI, Faridabad	November 2021	May 2022
Senior Research Scientist	THSTI, Faridabad	July 2021	November 2021
Scientist D	THSTI, Faridabad	January 2020	June 2021
Scientist C	THSTI, Faridabad	January 2019	January 2020
Early Career Medical Research Awardee	THSTI, Faridabad	July 2016	January 2019

D. Funded projects (as PI / Co-PI)

Title	Funding agency	Role	Duration / status	Amount
IPD analysis of hypothyroidism in pregnancy: prevalence,	Indian Council of Medical Research (ICMR), National Health	PI	20 August 2026 to 19 February 2028 (18 months)	INR 24,51,020 (Year 1 sanctioned)

risk factors and maternal and fetal outcomes in India (ePMS NHRP-2026-0000226)	Research Priority			
Secondary data analysis of gestational weight gain and maternal-neonatal outcomes in Indian cohorts (ePMS NHRP-2026-0000219)	Indian Council of Medical Research (ICMR), National Health Research Priority	PI	1 June 2026 to 30 November 2027 (18 months)	INR 19,07,060 (Year 1 sanctioned)
Surveillance and Pregnancy/Perinatal Research Platform for India: pregnancy risk stratification data analysis, modelling and capacity building (Research Service Agreement, THSTI P609)	Christian Medical College (CMC) Vellore	PI	1 October 2025 to 31 March 2028	INR 1,14,88,500 (excl. GST)
Graded Risk-based Assessment for Screening and Prevention of Placental Disorders (GRASPP): randomised controlled trial of a two-step contingent screening strategy for the great obstetrical syndromes (EMTR/IM-DEL/2025/03-28)	Indian Council of Medical Research (ICMR)	Co-PI	11 November 2025 to 10 November 2029	INR 9,94,20,000 (994.20 lakh)
MOMI-India: Multi-Omics for Mothers and Infants, Genomics and Analysis Planning, Phase II (BIRAC/GCI/2023/MOMI-Phase II)	Bill & Melinda Gates Foundation, via Grand Challenges India-BIRAC	Co-PI	23 August 2023 to 22 August 2027	INR 10,65,90,000 (1065.90 lakh)
Improving maternal and neonatal outcomes using imaging data science (IIRPIG-2023-0001097; File No. EM/Dev/IG/0001097/2023/JJC; PI: Dr Nitya Wadhwa)	ICMR, Extramural Research Programme (Intermediate Grant)	Co-PI (clinical investigation)	20 March 2024; approved duration 4 years, annual extension after review	INR 2,06,22,808 (Year 1 sanctioned); INR 7,97,46,000 proposed for 4 years
Effects of extreme heat on maternal, placental and fetal physiology, lactation and newborn health in India (HiP-India)	Wellcome Trust, UK	Co-PI (clinical domain)	2024 to 2029 (dates to be confirmed)	GBP 2 million (consortium total)
AI for Ultrasound Programme (AI4USG): multicentric evaluation of Indian population-specific tools for antenatal gestational age estimation (MEGA study), with THSTI and IIT Madras	Grand Challenges India (BIRAC - Gates Foundation)	PI (technical domain), validation lead	2025 to 2026	Approx. INR 8 crore (incl. supplementary grant)

Completed project	**Funding agency**	**Role**	**Duration**	**Amount**
Inter-Institutional Programme for Maternal, Neonatal and Infant Sciences: a translational approach to studying preterm birth (GARBH-Ini Phase 2)	DBT, Government of India	Co-Investigator (clinical and epidemiology domain)	2020 to 2025	INR 48,85,32,000 (4885.32 lakh)
GARBH-Ini Phase 2, THSTI component	DBT, Government of India	Co-PI (clinical and epidemiology domain)	2020 to 2025	INR 29,35,56,000 (2935.56 lakh)
A bench-to-bedside	DBT, Government of	Co-PI (clinical and	2018 to 2023	INR 6,81,90,000

model for clinical and translational science focused on fetal growth restriction and preterm birth (Glue Grant)	India	epidemiology domain)		(681.90 lakh)
Multi-Omics for Mothers and Infants (MOMI) Consortium, Phase I	Bill & Melinda Gates Foundation, via Grand Challenges India-BIRAC	Co-PI	2019 to 2023	INR 5,16,77,782 (516.78 lakh)
COVID-19 Bioresource at the NCR Biotech Science Cluster / DBT Consortium for COVID-19 Research	DBT, Government of India	Co-PI (cohort domain)	2020 to 2023	-
Preterm birth risk prediction in pregnant women using machine learning models	Grand Challenges India (BIRAC - Gates Foundation)	Co-Investigator	2020 to 2022	-

E. Intellectual property and translation

Title	Application no.	Status	Role in invention
Garbh-Ini Ultrasound Image-based Gestational Age Estimation System (GAUGE)	202411027837	Filed 4 April 2024; awarded	Co-conceived the invention, designed and conducted the clinical study, and clinically evaluated the model.
A population-specific gestational dating system and a method thereof	202341061748	Filed 13 September 2023	Conceived the invention, designed the clinical study, clinically phenotyped the participants from whom the model was developed, and clinically evaluated the model.
Potently neutralising novel human monoclonal antibodies to SARS-CoV-2 (COVID-19)	202111038519	Filed 25 August 2021; awarded	Designed the clinical study and clinically phenotyped the participants from whom the monoclonal antibodies were generated.
Immunoassay for detection of Coronaviridae	202011039696	Filed 14 September 2020; awarded	Designed the clinical study, clinically phenotyped the participants from whom biospecimens were collected, and clinically evaluated the diagnostic assay.

Translational partners: IIT Madras and the Robert Bosch Centre for Data Science and AI (AI for Ultrasound, GAUGE).

F. Five relevant publications

1. Thiruvengadam R, Ayushi, Murugesan DR, et al; GARBH-Ini Study Team. Incidence of and risk factors for small vulnerable newborns in north India: a secondary analysis of a prospective pregnancy cohort. *Lancet Glob Health*. 2024;12(8):e1261-e1277. doi:10.1016/S2214-109X(24)00212-2 *[First author]*

2. Gadekar VP, Damaraju N, Xavier A, et al; GARBH-Ini Study Group. Development and external validation of Indian population-specific Garbhini-GA2 model for estimating gestational age in second and third trimesters. *Lancet Reg Health Southeast Asia*. 2024;25:100362. doi:10.1016/j.lansea.2024.100362 *[Corresponding author]*

3. Thiruvengadam R, Awasthi A, Medigeshi G, et al; DBT India Consortium for COVID-19 Research. Effectiveness of ChAdOx1 nCoV-19 vaccine against SARS-CoV-2 infection during the delta (B.1.617.2) variant surge in India: a test-negative, case-control study and a mechanistic study of post-vaccination immune responses. *Lancet Infect Dis*. 2022;22(4):473-482. doi:10.1016/S1473-3099(21)00680-0 *[First author]*

4. Thiruvengadam R, Rizvi ZA, Raghavan S, et al. Clinical and experimental evidence suggest omicron variant of SARS-CoV-2 is inherently less pathogenic than delta variant independent of previous immunity. *Eur J Med Res*. 2023;28(1):421. doi:10.1186/s40001-023-01373-3 *[First author]*

5. Chanian R, Thiruvengadam R, Desiraju BK. Prediction of spontaneous preterm birth in the second trimester: current models, limitations, and the way forward. *Indian J Pediatr*. 2026;93(8):852-858. doi:10.1007/s12098-026-06030-1 *[Second author]*

G. Awards

- Award for a PhD student for the Best Published Paper of 2021, THSTI Foundation Day (DBT) (15 July 2022)
- Best oral presentation (second prize), Workshop on Air Pollution and Health Effects Research in India, CAPHER-India (21-23 October 2021)
- Best PhD oral presentation (second prize), National Science Day, THSTI (28 February 2021)
- Dr M.K. Bhan Group Award for the most impactful collaboration, THSTI (as a member of the clinical team of the DBT Consortium for COVID-19 Research) (2020)
- Early Career Medical Research Award, THSTI, DBT, Government of India (maternal and child health) (2017)